SAR Target Zero-Shot Recognition with Optical Image Assistance

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2024 IEEE International Conference on Signal, Information and Data Processing (ICSIDP) | 979-8-3315-1566-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSIDP62679.2024.10868138

***Abstract*—Most synthetic aperture radar (SAR) target recognition (ATR) methods focus on closed set conditions, which assumes that the testing set shares the same label space with the training set. However, new targets inevitably appear in practical testing phase. Closed-set SAR target recognition methods will misclassify those unseen new targets as the known targets in training phase, causing decision errors. Zero-shot recognition (ZSR) is an effective way to make predictions for unknown classes. Existing ZSR methods, which are primarily designed for optical images, typically construct an attribute knowledge base shared among known and unknown targets, and then learn the feature-attribute mapping using only known class data, which is subsequently applied to predict attributes of unknowns during testing and achieve class judgment. Unfortunately, the optical ZSR methods do not perform well in SAR data, due to the huge difficulty in the learning of feature-attribute mapping caused by the less accurate target representation of SAR images than the optical images. To overcome this problem, this paper proposes a ZSR method of SAR target with optical image assistance. In detail, on the basis of attribute knowledge base construction, the proposed method maps the known optical and SAR training data to the feature space through their respective feature extraction network, and uses feature aggregation loss to ensure that optical and SAR image features from the same category are close to each other, while those from different categories remain far apart. Based on the alignment of optical and SAR features, the feature-attribute mapping is learned using both SAR and optical data, leveraging the advantages of optical images in the easier and more accurate learning of feature-attribute mapping to promote the learning of SAR images. In testing phase, the learned SAR feature extraction network and feature-attribute mapping are used to predict unknown targets' attributes. Ultimately, the unknown class prediction is realized by measuring the similarities between the predicted attributes and true attributes from all unknown targets which can be obtained according to the prior knowledge. Experiments on measured ship dataset demonstrate that our method achieves superior recognition performance compared to existing ZSR methods using only SAR data.**

Keywords—Synthetic aperture radar(SAR), automatic target recognition(ATR), zero-shot recognition(ZSR), optical image assistance

I. INTRODUCTION

Synthetic aperture radar (SAR) has been widely applied in civilian and military fields in recent years due to its all-weather and all-time capabilities. With the gradual advancement of SAR systems and imaging algorithms, an increasing number of high-quality SAR images have emerged, which has further promoted the research and application of SAR target recognition technology[1].

In the research field of SAR target recognition, early scholars extract handcrafted features from SAR images to describe targets, such as scale-invariant feature transform (SIFT), and further achieve category identification[2]. Li et al.[3] combine two kinds of features, i.e., the multi-level amplitude feature and the multi-level Dense Scale-Invariant Feature Transform (Dense-SIFT) feature to achieve higher recognition performance. With the quick development of deep learning, an end-to-end feature learning framework, convolutional neural network (CNN) has been the mainstream method for SAR automatic target recognition (ATR) due to its powerful feature representation ability and high recognition performance. Chen et al.[4] propose a new all-convolutional networks (A-ConvNets) to replace fully connected layers in CNN, and achieve higher recognition accuracy compared to traditional ConvNets. Lin et al.[5] develope a channel-wise convolutional structure to extract polarimetric information from SAR images, which is helpful for SAR recognition. Zeng et al.[6] introduce a stream-based self-attention mechanism to adaptively learn the attention distribution of the multi-streams multi-scale sematic features, further enhancing the performance of SAR-ATR system. Although existing SAR ATR methods have achieved high recognition accuracy on specific datasets, they are mostly based on the closed set assumption, which considers that the testing set shares the same label space as the training set. However, SAR target recognition technology is often aimed at non-cooperative targets. In non-cooperative target recognition tasks, it is difficult to obtain high-value targets in advance and establish a complete recognition template library. When faced with unknown targets that are absent during the training phase, traditional closed set recognition methods will forcibly classify them into a known category within the library, leading to serious decision errors.

Zero-shot recognition (ZSR) is considered as an effective way to recognize new unseen targets in open set conditions, thus receiving widespread attention from researchers. ZSR methods typically rely on prior knowledge of the target's shape, structure and other characteristics to select easily learnable and discriminative attributes to describe both known and unknown classes, then appropriate quantification techniques are used to determine the class attribute vectors for all categories, forming an attribute knowledge base. Then, the feature-attribute mapping is learned using training data from known classes. In the testing phase, based on the extraction of unknown target features, the learned feature-attribute mapping is used to predict the attribute vectors of unknown classes. Ultimately, the recognition of unknown classes is achieved by measuring the similarity between the predicted attribute vectors and the predefined attribute vectors of all unknown classes. Akata et al.[7] propose an attribute-based label-embedding (ALE) model for ZSR, which extracts attribute information from known class images by constructing a forward feature-attribute mapping, and the unknown recognition is achieved by the predicted attribute vectors for the unknown targets. Zhang et al.[8] propose a deep embedding model (DEM) based ZSR method and construct a reverse feature-attribute mapping, which maps attribute vectors to the feature space. Thus, the unknown target recognition is performed in feature space. Xu et al.[9] propose a ZSR framework that jointly learns global and local features and achieve higher ZSR performance. Existing ZSR methods are mostly designed for optical images, and there is limited research on ZSR for SAR targets. Wei et al.[10] construct an attribute knowledge base for vehicle targets from public SAR dataset Moving and Stationary Target Acquisition and Recognition (MSTAR). A reverse mapping, similar to DEM, is then performed to achieve the SAR ZSR task.

The above ZSR algorithms are primarily designed for optical image classification. The few existing SAR ZSR methods are merely direct applications or simple adaptations of existing optical ZSR methods. Compared to the optical ZSR problem, the SAR ZSR problem is far more challenging. Due to the unique imaging mechanism of SAR, SAR images are less accurate than optical images in representing target structures, textures, and other characteristics. This makes it challenging to extract target attribute-related information from raw SAR images, and learning the feature-attribute mapping is even more difficult. In order to improve unknown class recognition accuracy in SAR ATR, this paper proposes an **o**ptical **i**mage **a**ssisted **ZSR** method (OIA-ZSR). In general, on the basis of constructing an appropriate attribute knowledge base for both known and unknown targets based on their physical characteristics, we introduce optical data with the same categories as the known SAR data to assist in the feature-attribute mapping learning for SAR images. Specifically, the proposed method uses a dual-branch deep neural network to separately extract high-level features from known SAR and optical images, subsequently mapping them into the same feature space with the feature aggregation loss. In detail, this loss exploits contrastive learning to ensure that features of the same category from SAR and optical images are closely aligned with the predefined class prototype, while features of different categories from SAR and optical images remain distinct and far apart. Based on the common features extracted from optical and SAR data, the feature-attribute mapping is performed using both SAR and optical data, leveraging the accurate attribute representation of optical images to enhance the accuracy of attribute learning for SAR targets. In the testing phase, the extracted feature of SAR testing target is compared to the attribute embedding features using feature-attribute mapping learned in training phase, and recognition is then performed by similarity measurement. Experiments on measured SAR ship dataset demonstrate that our proposed method achieves higher accuracy in recognizing unknown categories compared to the ZSR method using only SAR data.

The main contributions of this work are summarized as follows.

1) We propose a ZSR method, i.e., OIA-ZSR for SAR target. On the basis of extracting common features from both SAR and optical images, the feature-attribute mapping is jointly learned using both SAR and optical images. By leveraging the precise attribute representation of optical images, the difficulty of attribute learning for SAR images is reduced. The proposed method significantly enhances the accuracy of SAR attribute learning, leading to higher recognition accuracy for unknown SAR targets compared to the ZSR methods using only SAR data.

2) For the application of ship ZSR task, we construct an appropriate attribute knowledge base combining the physical characteristics of ship targets for the first time. This provides a foundation for attribute learning and accurate recognition of unknown targets.

II. PROPOSED METHOD FOR SAR ZSR

Fig. 1 shows the flowchart of our proposed OIA-ZSR method. As shown in Fig.1, our proposed method consists of three stages: 1) Attribute Knowledge Base Construction, 2) Feature-Attribute Mapping Learning with Optical Image Assistance, and 3) Unknown SAR Target Recognition. In stage 1), an appropriate set of attributes is constructed for both known and unknown targets according to their prior knowledge on physical structure, shape, functionality and so on. Then, an attribute knowledge base is formed through the quantification of known and unknown targets' attributes according to their specific attribute characteristics. In stage 2), a dual-branch network is used to extract SAR and optical features from known classes separately, with SAR and optical features being aligned by a feature aggregation loss in a same feature space. Meanwhile, each known class attribute vector is then mapped to the feature space, and is constrained to be close to the corresponding SAR and optical features using attribute mapping loss. Finally, the feature-attribute mapping can be learned via above two losses in an end-to-end manner. In stage 3), by using the learned SAR branch feature extraction network and feature-attribute mapping, the category inference for unknown SAR targets can be achieved by measuring the distance between the testing SAR feature and all unknown classes attribute embedding features. The detailed procedures of three stages are described as follows.

A. Attribute Knowledge Base Construction

Aiming to SAR ship ZSL task, we construct an attribute knowledge base shared by both known and unknown classes for SAR ships.

To construct an attribute knowledge base, some representative attributes are first selected based on prior knowledge of both known and unknown targets to construct the attribute space, and then, based on the specific attribute characteristics of known and unknown targets, the class attribute vectors are quantified, forming the attribute knowledge base. Since ZSR methods mainly rely on attributes for unknown class recognition, the selection of representative attributes is crucial for ZSR. Selected attributes need to be discriminative, ensuring that all categories have different attribute values. Additionally, they should be visible from raw images, making learn the attribute related information possible. For SAR ship ZSR task, by crawling web pages and reviewing relevant literature, we choose length, goods, fish-gear, weapon, aspect ratio and number of hulls to form the attribute space. Specifically, in conjunction with the ZSR experiments, we define the attribute knowledge base for each category, as shown in Table 1.

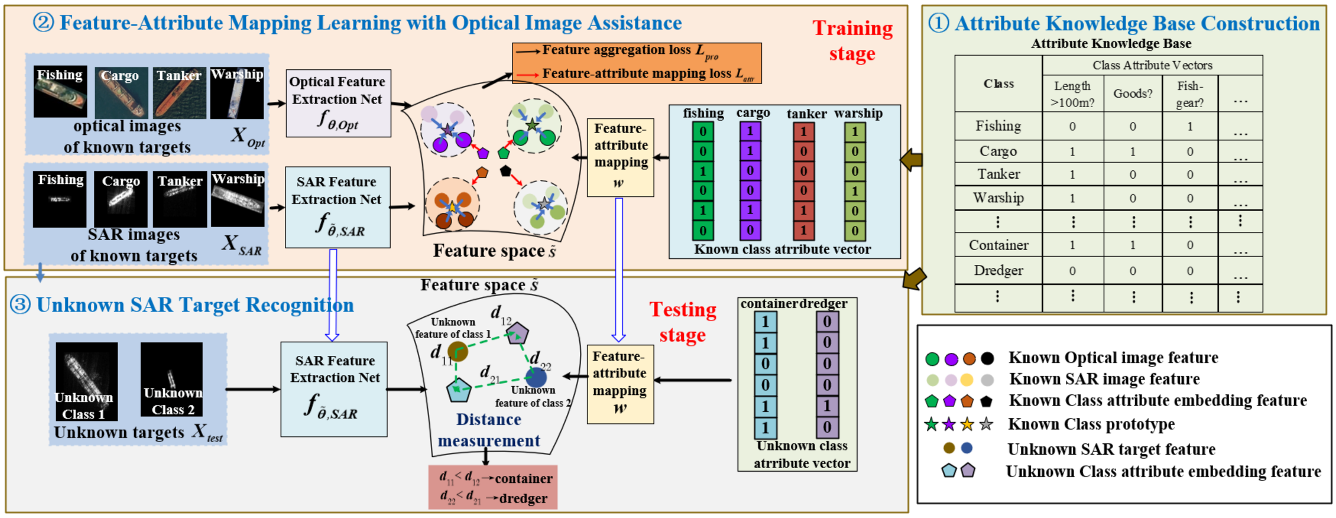


Fig. 1. Flowchart of the proposed OIA-ZSR method

TABLE I. ATTRIBUTE KNOWLEDGE BASE FOR SHIPS

Class	*Class Attribute Vectors*					
	Length >100m ?	*Goods?*	*Fish gear?*	*Weapo n?*	*aspect ratio <7?*	*Double hulls?*
Fishing	0	0	1	0	1	0
Cargo	1	1	0	0	1	0
Tanker	1	0	0	0	1	1
Warship	1	0	0	1	0	0
Contain er	1	1	0	0	1	1
Dredger	0	0	0	0	1	0

As Table I shows, we select six attributes for our ship ZSR task and use prior information on category attributes to determine the class attribute vectors through appropriate quantification methods. Specifically, for the attributes of fish gear, goods, and weapon, a value of 1 indicates the presence of the attribute, while 0 indicates its absence. For the length attribute, ships longer than 100 meters are assigned a value of 1, and those shorter than 100 meters are assigned a value of 0. For the aspect ratio attribute, ships with an aspect ratio less than 7 are assigned a value of 1, while those with an aspect ratio greater than 7 are assigned a value of 0. For the hull attribute, double-hulled ships are assigned a value of 1, and single-hulled ships are assigned a value of 0.

B. Feature-Attribute Mapping Learning with Optical Image Assistance

This stage mainly learns the feature-attribute mapping using both SAR and optical data from known classes. The details are as follows:

Given a SAR and optical training set $X_{SAR} = \left\{\left(x_{i,SAR}, y_i\right)\right\}_{i=1}^{N_{SAR}}$, $X_{Opt} = \left\{\left(x_{i,Opt}, y_i\right)\right\}_{i=1}^{N_{Opt}}$, where N_{SAR} and N_{Opt} represents the number of SAR or optical training data respectively, $x_{i,SAR}$ represents the ith SAR images, while $x_{i,Opt}$ represents the ith optical images, y_i denotes its corresponding label, the class labels y_i span from 1 to N_s ,where N_s denotes the number of classes for both SAR and optical training set. In this stage, we first use a dual-branch networks to extract high level information from SAR and optical raw images, denoted as $f_{\tilde{\theta},SAR}$ and $f_{\theta,Opt}$, and then feature aggregation loss is performed to constrain features of the same category from optical and SAR images to be close to each other, while ensuring that features of different categories are far apart. Specifically, we set a learnable prototype vector for each known class, denoted as $m_i, i = 1,2,\ldots N_s, m_i \in \mathbb{R}^d$ with d representing the feature dimension. The feature aggregation loss L_{pro} can be divided into two parts, representing the feature aggregation loss for SAR images $L_{pro,SAR}$ and optical images $L_{pro,Opt}$, respectively, which is formulated as:

$$L_{pro} = L_{pro,SAR} + \lambda L_{pro,Opt} \tag{1}$$

$L_{pro,SAR}$ can be expressed as:

$$L_{pro,SAR} = \sum_{k=1}^{N_s} \sum_{i=1}^{N_{k,SAR}} \frac{\exp\left(d\left(f_{\tilde{\theta},SAR}\left(x_{k,i,SAR}\right), m_k\right)\right)}{\sum_{k'=1,k'\neq k}^{K} \exp\left(d\left(f_{\tilde{\theta},SAR}\left(x_{k,i,SAR}\right), m_{k'}\right)\right)} \tag{2}$$

where $x_{k,i,SAR}$ represents the ith sample in the kth category of SAR ship training set, $N_{k,SAR}$ represents the total number of samples in the kth category of SAR images, $f_{\tilde{\theta},SAR}$ represents the nonlinear feature transformation of the SAR feature extraction branch with learnable parameter $\tilde{\theta}$, m_k represents the prototype of the kth category, and $d(x,y)$ denotes the distance between x and y . $L_{pro,SAR}$ constrains the extracted features to be close to their corresponding category prototypes and far from other category prototypes, thus achieving a learnable and separable feature space. Similarly, the feature aggregation loss for optical images $L_{pro,Opt}$ can be represented as:

$$L_{pro,Opt}=\sum_{k=1}^{N_s}\sum_{i=1}^{N_{k,Opt}}\frac{\exp\left(d\left(f_{\theta,Opt}\left(x_{k,i,Opt}\right),m_k\right)\right)}{\sum_{k'=1,k'\neq k}^{K}\exp\left(d\left(f_{\theta,Opt}\left(x_{k,i,Opt}\right),m_{k'}\right)\right)} \tag{3}$$

where $x_{k,i,Opt}$ represents the ith sample in the kth category of optical ship training set, $N_{k,\mathrm{Opt}}$ represents the total number of samples in the kth category of optical ship images, and $f_{\theta,Opt}$ represents the nonlinear feature transformation of the optical feature extraction branch with learnable parameter θ . λ in (1) adjusts the proportion of SAR and optical feature aggregation losses. These two parts collectively constrain the extracted features of optical and SAR images to be close to their corresponding category prototypes, thereby achieving cross-domain alignment of optical and SAR image features and a shared feature space for SAR and optical image features $\tilde{S}$ is formed. Afterwards, attribute mapping loss L_{attr} is exploited to learn the feature-attribute mapping $w: A\rightarrow\tilde{S}$,where A denotes the set of all known attribute vectors in knowledge base. In detail, L_{attr} can be described as follows.

$$L_{attr}=L_{attr,SAR}+\lambda L_{attr,Opt} \tag{4}$$

The attribute mapping loss L_{attr} can be divided into two parts, representing the loss from SAR images $L_{attr,SAR}$ and the loss from optical images $L_{attr,Opt}$, respectively, which are formulated as:

$$L_{attr,SAR}=\sum_{u\in X_{SAR}}L\left(w\left(A_{y(u)}\right),f_{\theta',SAR}\left(u\right)\right) \tag{5}$$

$$L_{attr,Opt}=\sum_{v\in X_{Opt}}L\left(w\left(A_{y(v)}\right),f_{\theta,Opt}\left(v\right)\right) \tag{6}$$

where $y(x)$ represents the corresponding label of sample x , $A_{y(x)}$ represents the class attribute vector for sample x , $w: A\rightarrow\tilde{S}$ represents the mapping from attribute space to feature space, and $L(x,y)$ denotes the loss between x and y . Here we use MSE loss, which is formulated as follows.

$$L(x,y)=\| x-y\|^2 \tag{7}$$

The total loss for network training is the sum of the feature aggregation loss and the attribute learning loss, described as

$$L_{tol}=L_{pro}+L_{attr} \tag{8}$$

The SAR and optical feature extraction networks, along with the feature-attribute mapping, are jointly optimized according to the loss in (8), ensuring that the learning of SAR feature-attribute mapping is influenced by the learning of optical feature-attribute mapping. Since optical image attributes are easier to learn and the extracted features accurately reflect target attributes, this effectively enhances the accuracy of attribute-related feature learning in SAR images. Ultimately, this guarantees the accuracy of the feature-attribute mapping learning.

C. *Unknown SAR Target Recognition*

This stage utilizes the SAR feature extraction network and feature-attribute mapping learned in section II-B to predict the categories of unknown SAR targets during the testing phase. The details are as follows.

To predict the label of test sample $\tilde{x}$ from unknown class, we first use SAR feature extraction network to get feature representation $f_{\tilde{\theta},SAR}\left(\tilde{x}\right)$, then we use w to map all unknown classes attribute vectors to feature space, denoted as $U=\left(U_1,U_2,...U_{N_u}\right)$, where N_u represents the total number of unknown categories, $U_k=w\left(A_k\right),k\in\{1,2,..N_u\}$ represents the kth mapped feature from its corresponding attribute vector. The prediction can be made as follows.

$$\hat{k}=\arg\min_{k}\| f_{\tilde{\theta},SAR}\left(\tilde{x}\right)-U_k\|^2 \tag{9}$$

III. EXPERIMENTS

A. *Experimental Dataset Description*

All experiments in this paper are based on our custom ship dataset, which comprises measured SAR ship images collected from GF-3 satellite and public FuSAR[11] dataset, and optical images from public dataset FAIRlM[12] and FGSCR[13]. The SAR ship dataset includes six categories: fishing, cargos, tankers, warships, containers, and dredgers, with quantities of 115, 233, 121, 51, 48, and 66 respectively. The optical ship dataset includes four categories: fishing, cargos, tankers, and warships, with quantities of 125, 173, 110, 51.

Fig. 2 provides examples of some SAR ship images and optical ship images from the dataset, respectively.

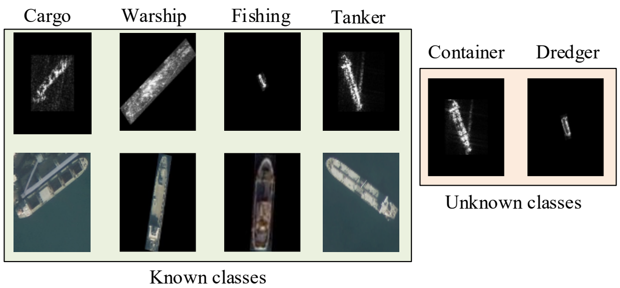


Fig. 2. Examples of different ships in the SAR and optical ship datasets

B. *Experiment Setup and Evaluation Metrics*

The experimental data are derived from our custom SAR and optical ship datasets, which have been described in detail in Section III A. In the experiment, we consider fishing, cargos, tankers, warships to be known classes, while the remaining containers and dredgers are considered as unknown classes. The hyper-parameter setup and running environment for the proposed network in the experiment are as follows:

λ in loss function (1) and (4) is set to be 0.7, the initial learning rate with Adam is set to 0.001 and decays by 0.50 every 50 epochs, and the total training epochs are set to 200. Our method is implemented by using the open framework PyTorch and on an AMD Ryzen 7 5800 CPU, and an NVIDIA GeForce RTX 3060 GPU.

In experiment, class recognition accuracy and average class recognition accuracy are selected as evaluation metrics. These metrics measure the recognition accuracy of each class and the average recognition accuracy of all classes, respectively. The calculation formulas are shown below:

$$Acc_{container/dredger}=\frac{c_{container/dredger}}{N_{container/dredger}} \tag{10}$$

$$Acc = \frac{1}{2}(Acc_{container} + Acc_{dredger}) \quad (11)$$

where $c_{container/dredger}$ represents the number of correctly classified samples for the container or dredger class, and $N_{container/dredger}$ represents the total number of samples in the container or dredger class. The average recognition accuracy Acc is the mean of the two unknown class recognition accuracy.

C. Comparison Experiment

This section will compare the proposed OIA-ZSR method with three other related methods, including ALE[7], DEM[8] and APN[9], in terms of the class recognition accuracy and the average recognition accuracy. The comparison methods mentioned above are designed for optical images. In this paper, we directly apply them to SAR image recognition for comparison. In detail, the ALE maps visual features to the attribute space and performs unknown class matching and recognition based on the predicted attributes of unknown classes. In contrast, the DEM method achieves unknown class recognition in the feature space by constructing a reverse feature-attribute mapping, which is essentially the same as our method without optical assistance. The APN method learns both global and local features, and achieve higher recognition accuracy. However, all of them use only SAR data for training, which limits the zero-shot recognition performance for unknown classes. Tables II shows the recognition results of the proposed method and the comparison methods.

As shown in Table II, our proposed method achieves the highest accuracy for both individual class recognition and average class recognition. Compared to the ALE method, the proposed method improves the recognition rate of container by 6.25%, and dredger by 4.55%. The average class recognition accuracy is improved by 5.40%. We consider there are two main reasons. On the one hand, as mentioned in [8], the reverse mapping from attribute to feature we use alleviates the hubness problem in ZSR, making recognition easier by performing recognition in a high-dimension space. On the other hand, the optical ship data we use during the training phase assist in the attribute learning process for SAR data, resulting in higher recognition performance for unknown classes, which also leads to the higher recognition accuracies than the DEM and APN.

TABLE II. RECOGNITION RESULTS FOR DIFFERENT ZSR METHODS

Methods	$Acc_{container}$ ↑	$Acc_{dredger}$ ↑	Acc ↑
ALE[7]	93.75%	92.42%	93.09%
DEM[8]	95.83%	95.45%	95.64%
APN[9]	95.83%	**96.97%**	96.40%
Proposed method	**100%**	**96.97%**	**98.49%**

D. Rationality Analysis of Attribute Knowledge Base

As mentioned in INTRODUCTION, we first establish the ship attribute knowledge base for SAR ZSR study. This section will analyze the rationality of the constructed ship attribute knowledge base.

TABLE III. RECOGNITION RESULTS WHEN REMOVING ATTRIBUTE

Removed Attribute	$Acc_{container}$ ↑	$Acc_{dredger}$ ↑	Acc ↑
Length	87.50%	90.91%	89.21%
Good	83.33%	87.88%	85.61%
Fish gear	97.92%	95.45%	96.69%
Weapon	97.92%	95.45%	96.69%
aspect ratio	95.83%	93.94%	94.89%
hulls	97.92%	93.94%	95.93%
All attributes	**100%**	**96.97%**	**98.49%**

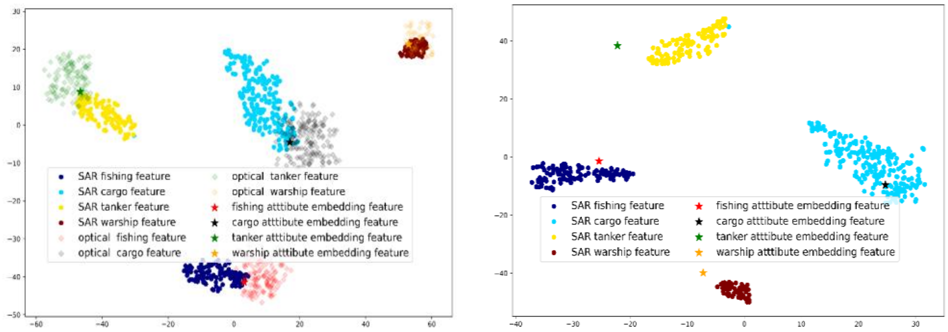
In section II-A, we select six attributes for ship ZSR. To validate the rationality of the constructed attribute knowledge base, we analyze each attribute's impact by sequentially reducing the attribute space dimensions. This involves removing one attribute at a time while retaining the remaining five attributes, and conducting ZSR experiments using the reduced attribute space. Table III presents the recognition accuracies for container and dredger after sequentially removing each attribute.

From Table III, it is evident that the removement of any attribute leads to a decrease in both individual class recognition accuracy and average recognition accuracy, reflecting the rationality of the six-dimensional attribute space established in this paper. Additionally, it can be observed that when the attributes of length and good are reduced, the recognition accuracy for container and dredger significantly decreases, indicating the high importance of these two attributes in distinguishing these two unknown classes.

E. Effectiveness Analysis of optical image assistance strategy

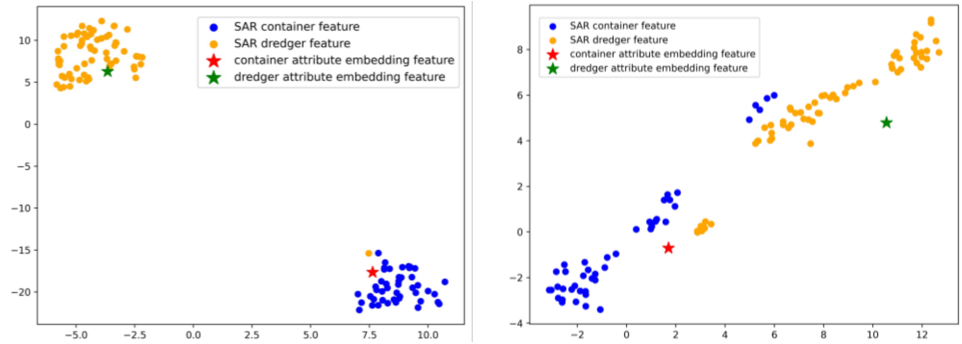
To verify the effectiveness of optical image assistance, Fig. 3(a) and 3(b) show the t-Distributed Stochastic Neighbor Embedding(t-SNE) visualization of input training images after network training convergence with or without optical image assistance.

As shown in Fig. 3(a), after training convergence, we find that the optical image features and SAR image features of the same category achieve cross-domain alignment, demonstrating the effectiveness of the feature aggregation loss. Meanwhile, the optical and SAR features from the same category cluster around the corresponding class attribute embedding vector. In contrast, in Fig 3(b), the SAR features have a noticeable distance from the corresponding category's attribute embedding feature. This indicates that when using only SAR data for attribute learning, it is challenging to accurately learn the feature-attribute mapping relationship. However, with the introduction of optical image, the SAR features are closer to the corresponding category attribute embedding features, demonstrating that optical images indeed assist in the SAR attribute learning process, leading to a more accurate feature-attribute mapping.



(a) t-SNE visualization results of training set with optical image assistance (b) t-SNE visualization results of training set without optical image assistance

Fig. 3 t-SNE visualization results of training data from four known classes in the feature space with or without optical image assistance.



(a) ZSR with optical image (b) ZSR without optical image assistance

Fig. 4 t-SNE visualization results of the two unknown classes in the feature space with and without optical image assistance.

Furthermore, we present the visualization of two unknown class features during the testing phase, with or without optical assistance in Fig 4 (a) and (b).

From Figure 4(a) and (b), it is evident that the introduction of the optical assistance strategy results in a more concentrated distribution of features for the two unknown classes, bringing them closer to the corresponding attribute embedding features. This indicates that the optical assistance strategy indeed enhances the accurate representation of features to attributes, thereby improving the recognition performance of unknown class.

IV. CONCLUSION

To address the issue of difficulty for SAR ZSR in the accurate learning of feature-attribute mapping, this paper proposes a SAR zero-shot recognition method with optical image assistance. The proposed method first constructs an attribute knowledge base for all known and unknown targets according to their prior knowledge, and then feature mapping loss is used to learn feature-attribute mapping with both known SAR and optical data based on the alignment of SAR and optical features caused by the feature aggregation loss. By leveraging the advantages of optical images in the more accurate representation of target and easy-to-learn feature-attribute mapping, our method enhances the attribute learning capability of SAR data and improves the recognition accuracy for unknown classes. Experiments on measured SAR ship dataset have validated the rationality of the constructed ship attribute knowledge base and the effectiveness of the proposed optical assistance strategy. Our method achieves higher recognition accuracy than existing ZSR methods using only SAR data.

ACKNOWLEDGMENT

This research is funded in part by the National Science Foundation of China (No.U21B2039,No.62201433)， in part by the Fundamental Research Funds for the Central Universities(QTZX23067),and in part by the 111 Project (B18039).

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